

Department of Electronics Engineering
B.Tech. III Sem. V

Mid-Semester Examination, September – 2025

Digital Signal Processing (EC303)

Instructions:

1. Attempt all questions.
2. Assume necessary data if required.
3. Total number of questions are 5.

Time: 11:15 AM

Duration: 1 hour 30 minute.

Total Marks: 30

Q-1	A discrete-time LTI system is described by the difference equation: $y(n) - \frac{1}{2}y(n-1) = x(n)$, where $x(n) = \{1, 2, 3, 4\}$ for $0 \leq n \leq 3$. (i) Calculate the magnitude and phase of the transfer function. (ii) Compute the magnitude and phase of the output signal $y(n)$.	(07)
Q-2	Given, a signal $x(n]$. It's sample values are $x(0)=-2$, $x(1)=1$, $x(2)=1$ and $x(3)=2$. The discrete Fourier transform (DFT) of the signal is given by $X(k)$. Calculate the value of $X(0)$, $X(1)$, $X(2)$, $X(3)$, $X(4)$, $X(5)$, $X(6)$ and $X(7)$.	(06)
Q-3	A signal $x(n]$ is given by $x(n) = a^n u(n+2)$, and another signal $y(n]$ is related with $x(n]$ by following equation $y(n) = x(\frac{1}{2}n+2)$. Calculate the Z-transform of $y(n]$. Also mention the ROC in all cases.	(05)
Q-4	Explain the Decimation-in-frequency radix-2 FFT algorithm for 16-point DFT calculation. Also draw the butterfly diagram with detail explanation of each step.	(05)
Q-5	A pressure gauge that can be modelled as LTI system has an output $(1 - e^{-t} - te^{-t})u(t)$ to a unit step input. For a certain input $x(t)$, the output observed is $(2 - 3e^{-t} + e^{-3t})u(t)$. What should be the input for that, calculate it.	(07)

----- ALL THE BEST -----